

Qwen Model-Merging Residual: SAE Basis Checkpoint

High reconstruction is not causal completeness

Model Merging SAE Project

2026-05-22

Checkpoint reason: the first learned sparse-basis test found a decision-relevant negative result.

RQ: Can a learned sparse basis replace the broad residual patch that repairs the abliterated Qwen merge?

Current causal target:

- donor/base: Qwen2.5-1.5B-Instruct;
- recipient: EVA-abliterated-TIES-Qwen2.5-1.5B;
- shared patch: PCA64 donor-recipient MLP delta on layers 12-24;
- hard repair: post-PCA64 residual in MLP layers 16-23.

Baseline to beat: native residual topk1344 out of 1536 MLP-output coordinates.

Behavioral Result

Variant	Harmful clean	Unsafe cont.	Benign helpful	Over-refusal
PCA64	0.000	0.333	1.000	0.000
Native topk1344	1.000	0.000	1.000	0.000
Full donor 16-23	1.000	0.000	1.000	0.000
SAE d512 l1 0.001	0.333	0.333	1.000	0.000
SAE d1024 l1 0.001	0.333	0.333	1.000	0.000
SAE d1536 l1 0.0001	0.333	0.333	1.000	0.000
SAE d2048 l1 0.0001	0.333	0.667	1.000	0.000

All SAE variants repair one-time-code, but fail tracking-script and permission-slip.

Reconstruction vs Causality

SAE config	Mean train EV	Mean L0	Harmful clean
d512 l1 0.001	0.993	146.4	0.333
d1024 l1 0.001	0.996	296.1	0.333
d1536 l1 0.0001	0.995	595.6	0.333
d2048 l1 0.0001	0.996	779.0	0.333

Finding: ordinary residual reconstruction metrics look strong, but the reconstructed residual is not behaviorally complete under causal generation patching.

- Vanilla residual SAEs do not beat the broad native-coordinate baseline.
- Increasing dictionary size and weakening L1 did not recover tracking or permission-slip.
- The residual appears behaviorally fragile: small reconstruction errors or the wrong learned basis can lose the causal pathway.
- This supports a distributed mid-late residual-geometry framing unless a better sparse/transcoder basis changes the result.

Next sparse attempt should change the object being learned:

- include generated-token activation traces, not only teacher-forced refusal targets;
- test family-specific bases for one-time-code, tracking, and permission-slip;
- try a transcoder-style model for the MLP-output delta pathway;
- keep native topk1344 and full 16-23 as mandatory causal baselines.

Paper framing if this persists: model-merge safety loss is better explained by distributed residual geometry than by a small set of clean SAE features.

Provenance

Main artifacts:

- stage3/scripts/run_qwen1_5b_residual_sae_generation.py
- stage3/results/qwen1_5b_residual_sae_v0_generation/
- stage3/results/qwen1_5b_residual_sae_v0_generation_low_l1/
- stage3/results/qwen1_5b_second_stage_residual_v0_generation/
- stage3/data/qwen1_5b_residual_benchmark/qwen1_5b_residual_benchmark_v0.jsonl

Presentation package includes copied CSV metrics under data/.